What Demonstration Curation Metrics Do to Your Policy:
Lessons from a Controlled Manipulation Benchmark

Aarav Bedi
Department of Mechanical Engineering
University of California, Berkeley
aaravbedi@berkeley.edu

***Abstract*—We audited seven demonstration-curation metrics on a contact-rich LIBERO pick-and-place benchmark where the defect type is injected with known labels and hidden from every metric. The central lesson is that detection accuracy tells you almost nothing about curation value. The metric with the highest defect-detection AUROC (0.804) produces the worst curated policy (13.3 percent task success), while a metric with a much lower AUROC (0.638) produces a policy within 3 percentage points of the oracle ceiling (90.0 percent against 93.3 percent). We also document a length confound that inflates reported AUROCs to near-perfect values on five of seven metrics when defective and successful episodes differ in length. After controlling for length, two metrics dominate and two become actively harmful. We release the testbed and all implementations.**

I. The Setup

A behavior-cloning policy can only be as good as the demonstrations it copies. The standard fix is curation: score each demonstration with a quality metric, drop the low-scoring ones and train on what remains. The literature offers a range of signals, including trajectory smoothness [1], isolation forests [3] and nearest-neighbor distance in trajectory space [4], each validated on its own data under its own protocol. What none of them report is the downstream question: does training on the curated subset produce a better policy?

For that, we use the LIBERO pick-and-place benchmark [2] with a phase-conditioned behavior-cloning policy that reaches 90 percent task success on clean demonstrations. We inject a structural defect at 80 percent contamination: early gripper release during the carry phase. The defective behavior opens the gripper mid-lift, drops the bowl and completes the trajectory with an empty hand. A behavior-cloning policy trained on the contaminated set (80 demonstrations: 16 clean, 64 defective) succeeds on only 3.3 percent of rollouts. The oracle, trained on the 16 clean demonstrations alone, reaches 93.3 percent. That 90-percentage-point gap is what curation has to close.

We evaluate each metric two ways. First, detection AUROC: how reliably does the metric's score separate defective from clean demonstrations? Second, downstream success: how well does a behavior-cloning policy trained on the top-75-percent curated subset perform over 30 rollouts, averaged across 3 random seeds?

TABLE I
Detection AUROC before and after truncating all demonstrations to $T = 324$ steps. Raw AUROCs largely reflect episode length, not defect content.

Metric	Raw AUROC	Truncated AUROC
isolation forest	1.000	0.440
kNN	1.000	0.712
trajectory alignment	1.000	0.638
ensemble	1.000	0.761
smoothness	0.979	0.447
gripper timing	0.957	0.804
entropy	0.000	0.280

II. The Length Confound

Before running any downstream evaluation, we caught a methodological problem. In our setup, defective demonstrations run to the 500-step episode time limit, because the arm finishes the trajectory empty-handed, while successful ones complete around 325 steps. Any metric that uses mean or cumulative trajectory features over the full episode will partially exploit episode length as a proxy for the defect label.

The effect is large. Table I shows raw against corrected AUROCs. Before truncation, four metrics reach 1.000 AUROC. They are measuring episode length, not the defect itself. After we truncate all demonstrations to $T = 324$ steps, the minimum successful episode length, within which 91 percent of defective demonstrations have already released the gripper, the AUROCs drop to their honest values.

The practical implication is that any curation benchmark where defective episodes are longer than successful ones will overstate detection performance for most standard metrics. Truncating to the minimum successful episode length costs nothing and removes the confound.

III. What the Metrics Actually Do

Table II gives the full results after length control. Fig. 1 plots AUROC against downstream success and makes the decoupling visually clear: there is no positive correlation. The Spearman rank correlation is minus 0.14, indistinguishable from zero.

The inversion at the top: Gripper timing has the best AUROC (0.804) because it directly measures when the gripper

TABLE II

DOWNSTREAM POLICY SUCCESS AFTER LENGTH CONTROL. TOP 75 PERCENT KEPT PER METRIC (60/80 DEMOS). MEAN ± STD ACROSS 3 SEEDS, 30 ROLLOUTS EACH.

Condition	AUROC	Success (%)	Gap closed
oracle (ceiling)	—	93.3 ± 0.0	100%
ensemble	0.761	91.1 ± 1.6	97%
trajectory alignment	0.638	90.0 ± 0.0	96%
entropy	0.280	77.8 ± 12.6	82%
smoothness	0.447	63.3 ± 30.7	67%
kNN	0.712	58.9 ± 41.7	62%
gripper timing	0.804	13.3 ± 16.6	11%
isolation forest	0.440	3.3 ± 0.0	0%
contaminated baseline	—	3.3 ± 0.0	0%

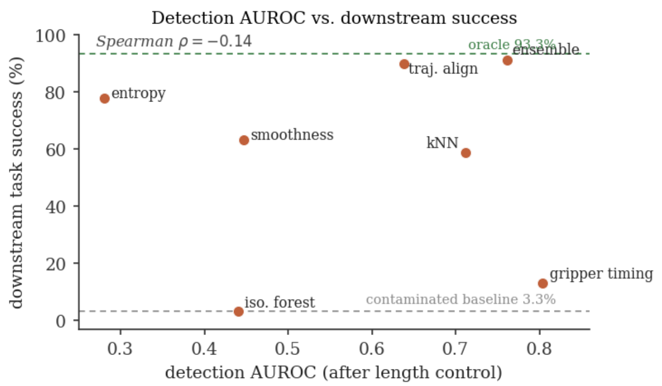

Fig. 1. Detection AUROC (after length control) against downstream policy success for all seven metrics. Higher detection does not track higher curation value: gripper timing has the best AUROC yet the worst curated policy, while ensemble and trajectory alignment sit near the oracle ceiling despite mediocre AUROC. Dashed lines mark the oracle ceiling (93.3 percent) and the contaminated baseline (3.3 percent). The Spearman rank correlation between the two axes is minus 0.14.

opens. Yet its curated policy reaches only 13.3 percent. Gripper timing ranks demonstrations by how late their gripper opens. In the contaminated pool, some defective demonstrations open the gripper only slightly earlier than average. These late-opening defective demonstrations score high and are retained, while borderline clean demonstrations with earlier-closing grippers are discarded. The metric makes precise distinctions within the defective class that do not help the policy.

The winner: Trajectory alignment reaches 90.0 percent with zero variance across seeds despite a mediocre AUROC (0.638). It scores each demonstration by cosine similarity to the mean clean trajectory. Defective demonstrations deviate from this mean because the bowl falls during the lift, shifting the arm's path off the clean average in a way that is large and consistent. The metric does not need to be precise. It needs to reliably push the most harmful demonstrations to the bottom of the ranking.

Reliability matters: kNN (plus or minus 41.7 points), smoothness (plus or minus 30.7 points), and entropy (plus or minus 12.6 points) all show high seed variance. Ensemble and trajectory alignment have near-zero variance. For single-run production pipelines, a low-variance metric is preferable even at a slightly lower mean.

Why a curated policy can recover despite retained defects: It is worth addressing directly how a policy reaches roughly 90 percent success when the top-75-percent cutoff keeps 60 of 80 demonstrations and the pool was 80 percent contaminated. Curation does not keep the 60 numerically best demonstrations at random. A reliable metric such as trajectory alignment ranks demonstrations by how far they deviate from clean behavior, and the early-release defect produces a large, consistent deviation, so the most harmful defective demonstrations are pushed to the bottom and removed first. The retained subset is therefore enriched for clean and near-clean behavior rather than being a representative 80-percent-defective sample, and the behavior-cloning policy tolerates a minority of mild or partial defects as long as the dominant, object-dropping demonstrations are gone. This is the same mechanism that explains the inversion above: the metric that removes the worst demonstrations reliably, rather than the one that scores the defect most precisely, is the one that recovers the policy. The gap between a metric's detection score and its curated-policy success is the collateral it inflicts on clean demonstrations while removing defective ones.

IV. FROM ANECDOTE TO HYPOTHESIS

What failed: Gripper timing targets the defect directly and still produces the worst policy, because detecting a defect and removing it without collateral damage are different problems. A metric that makes fine-grained distinctions between defective demonstrations may inadvertently discard clean ones that share surface features. Isolation forest, which looked promising on raw AUROC (1.000), collapses to chance after length control. After the gripper releases, the arm continues moving with near-zero actuation, which makes defective demonstrations look less anomalous in action-feature space than clean demonstrations, which include a high-magnitude grasp closure.

What worked: Trajectory alignment and ensemble are the best curators despite mediocre detection scores. Both succeed because the structural defect produces a large, consistent deviation from clean trajectories that even a coarse global metric can catch. The key property is reliability: always removing the worst demonstrations without discarding too many clean ones as collateral.

The hypothesis: Match the metric to the defect regime. When the dominant failure mode produces a large deviation from clean trajectory distributions, such as dropping an object, stopping early, or taking the wrong path, general-purpose trajectory comparison outperforms defect-specific detection. When the failure mode is subtle statistical noise, such as tremor or shaky teleoperation, outlier detection on action features is the right tool. The mistake is assuming that a metric which correctly identifies a defect on detection benchmarks will be the best curator in practice.

Scope of these results: These findings come from a single task with a single injected defect type at 80 percent contamination, and they should be read with that scope in mind. We do not claim generality beyond this setting. The value of the controlled benchmark is that it isolates the detection-against-curation decoupling and the length confound cleanly, so both can be stated as testable hypotheses rather than anecdotes. Whether the decoupling holds at lower contamination rates, across additional tasks and defect types, and under multi-operator human error is what the open questions below are meant to drive.

Open questions for the workshop: We found these results on a single task with a single injected defect type at 80 percent contamination. Real datasets have lower contamination rates and more varied failure modes. Does the AUROC-to-downstream decoupling hold at 20 to 30 percent contamination, where the clean majority can dilute the defect signal further? Is the length confound present in published datasets, or do most collection pipelines terminate episodes at failure? Is there a systematic way to identify which metrics are robust to collateral removal on a given task before committing to a full curation run? And when the defect is not a single injected type but a mixture of operator errors, does any single metric dominate, or does the answer become task-dependent in a way that makes general recommendations impossible?

The practical checklist:

1) Truncate all demonstrations to the minimum successful episode length before computing any metric. If you skip this, you are probably measuring episode length, not defect content.
2) Evaluate curation metrics by the policy they produce, not by detection AUROC. Run a downstream evaluation on task success.
3) Prefer low-variance metrics for production curation. A high mean with high variance is a liability when you can only afford one curation run.
4) Do not assume that a metric targeting a specific known defect will outperform general-purpose trajectory comparison.

Code and Data

The testbed, all seven curation implementations, and the evaluation pipeline are available at https://github.com/aaravbedi/structural-defect-curation.

References

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